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Chapter 7 Exercise 2:

A gas in thermal equilibrium at a temperature T has the velocity distribution

$$\rho(v)\, dv = \frac{m}{2\pi\tau}^{\frac{3}{2}} * 4\pi v^2 * e^{\frac{-mv^2}{2\tau}} dv$$

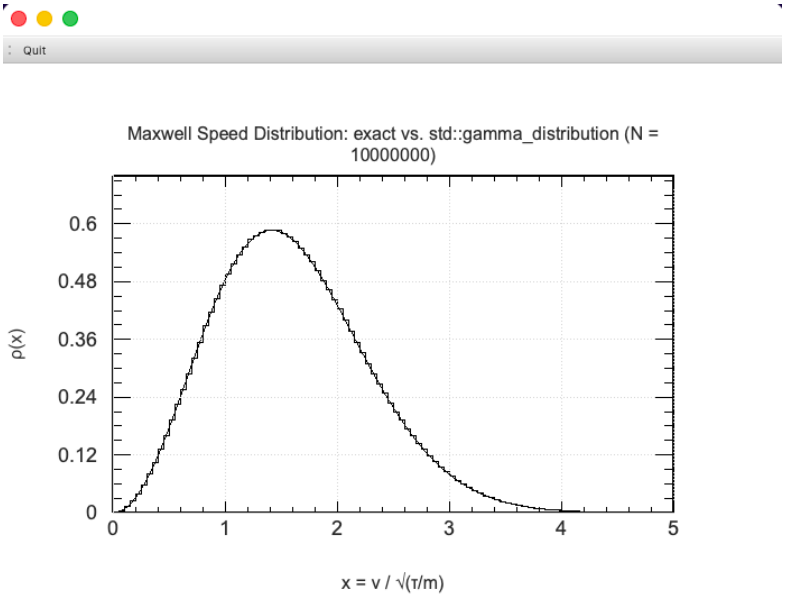
where $\tau = k_B T$, m is the mass of a molecule, and k_B is the Boltzmann constant. Nondimensionalize this and show that the Maxwell velocity distribution can be sampled with the aid of a gamma distribution. Make a plot of the velocity distribution together with data generated using the std::gamma_distribution of the standard library.

Solution

I created a double method for the distribution, promoted it to an F1D and then plotted it for the curve.

I then created a Hist1D for the histogram plot, and I used N = 10000000 samples across 100 buckets. I then sampled from the std::gamma_distribution using the random number generator. Then I ran a loop over N to do the sampling and normalized using sqrt(2u) and plotted the accumulated histogram against the curve. Increasing N promotes accuracy but, in turn, costs significantly more computing time.

Result



Chapter 7 Exercise 3:

Repeat the previous problem, but this time generate the Maxwell distribution using Markov chain Monte Carlo. Plot the distributions and check that they agree with the theoretical line shape.

Method

The target distribution is

$$\rho(x) = \sqrt{\frac{2}{\pi}} x^2 e^{-\frac{x^2}{2}}, \quad x > 0$$

Instead of drawing independent samples, I used the Metropolis algorithm, which builds a random walk (a Markov chain) whose points are distributed according to ρ(x). From the current point x a new point is proposed by a uniform random step:

$$x' = x + \delta, \quad \delta \in [-s, s], \quad s = 1$$

The proposal is accepted with probability

$$A(x \to x') = \min\left(1, \frac{\rho(x')}{\rho(x)}\right)$$

If the move is rejected, the chain stays at x and x is counted again. Proposals with x' ≤ 0 are always rejected because ρ = 0 there. Only the ratio of densities is needed, so the normalization constant $\sqrt{(2/\pi)}$ cancels:

$$\frac{\rho(x')}{\rho(x)} = \left(\frac{x'}{x}\right)^2 e^{-\frac{x'^2 - x^2}{2}}$$

Solution

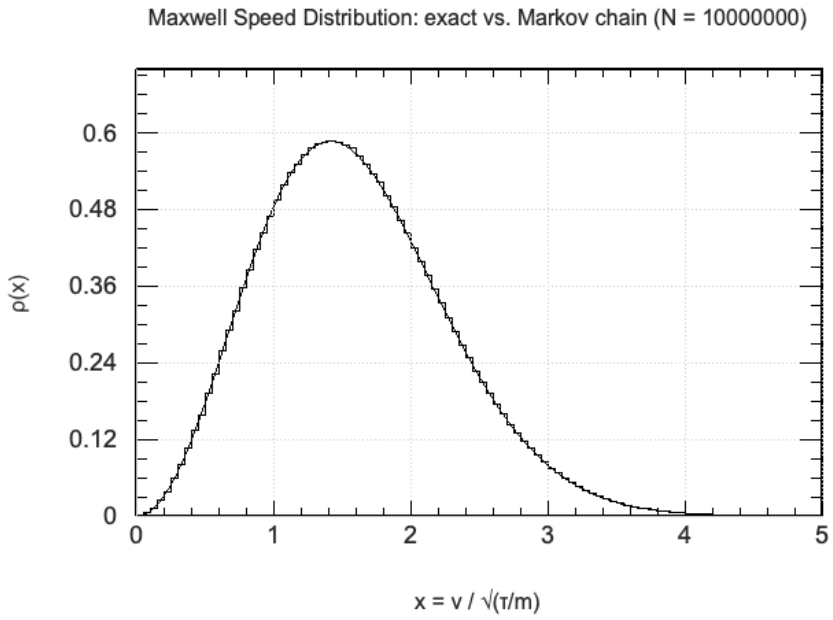
I reused the distribution function, the F1D curve, and the Hist1D/PlotHist1D setup from Exercise 2, and replaced the gamma sampling with a Metropolis loop. The chain starts at x = 1 (where ρ > 0). Each step proposes x' = x + δ using std::uniform_real_distribution on [−1, 1], computes the ratio ρ(x')/ρ(x), and accepts if a second uniform number on [0, 1) is smaller than the ratio. The current x is added to the histogram on every step, whether or not the move was accepted, since a rejection means the chain spent another step at x.

The first 1000 steps are ignored as burn-in so the starting point does not affect the result. I then recorded N = 10000000 steps in 100 bins on [0, 5], each with weight

$$w = \frac{1}{N\,\Delta x}$$

so the histogram has unit area and can be compared directly with ρ(x). The measured acceptance rate was about 72%.

Result



Chapter 7 Exercise 6

A particle that is polarized along the z direction decays to a pair of lighter particles, of which one is visible. The angular distribution of the visible particle is given by:

$$\rho(cos\theta) = (\frac{\sqrt{3}}{\sqrt{8}}P_1(cos\theta) - \frac{\sqrt{7}}{\sqrt{6}}P_3(cos\theta) + \frac{\sqrt{11}}{\sqrt{24}}P_5(cos\theta) - \frac{\sqrt{15}}{\sqrt{6}}P_7(cos\theta))^2$$

Generate data according to this distribution and plot it vs. cosθ in a histogram overlaid with the original function.

Method

Writing c = cos θ, the P_l are Legendre polynomials ($P_1(c) = c$):

$$P_3(c) = \frac{5c^3 - 3c}{2}$$

$$P_5(c) = \frac{63c^5 - 70c^3 + 15c}{8}$$

$$P_7(c) = \frac{429c^7 - 693c^5 + 315c^3 - 35c}{16}$$

Because the Legendre polynomials are orthogonal,

$$\int_{-1}^{1} P_l(c)P_{l'}(c)dc = \frac{2}{2l+1}\delta_{ll'}$$

There is no simple closed-form way to invert this distribution, so I used the accept–reject (rejection) method. A point is drawn uniformly inside the box that encloses the curve,

$$c \in [-1, 1], \quad y \in [0, \rho_{\max}]$$

$$accept\ c\ if \quad y < \rho(c)$$

and otherwise it is discarded.

Solution

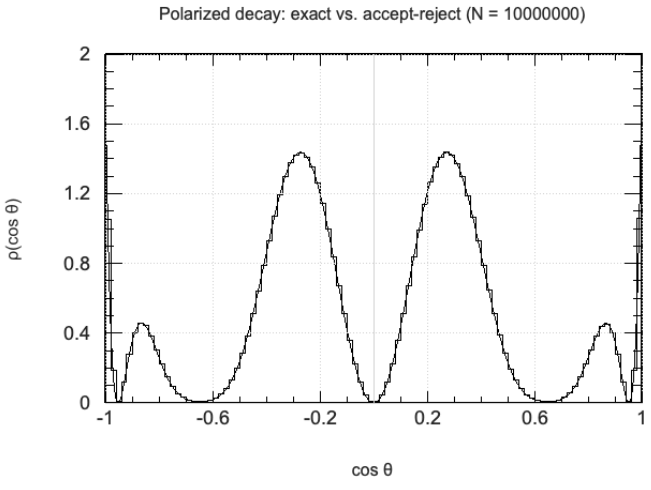
I wrote P1, P3, P5 and P7 as explicit polynomial functions and combined them in a distribution function that returns the square of the weighted sum; it is promoted to an F1D and plotted as the curve. To get ρ_max, the code scans ρ on a grid of 10001 points over [−1, 1] and adds a 1% safety margin. Two std::uniform_real_distribution objects then generate c on [−1, 1] and y on [0, ρ_max], and the loop keeps drawing until N = 10000000 values of c have been accepted.

Each accepted c is put in a Hist1D with 100 bins on [−1, 1], using the weight

$$w = \frac{1}{N\ \Delta c}$$

so the histogram has unit area and lies directly on top of ρ(c).

Result



Chapter 7 Exercise 12

In the transformation method (Section 7.2.7), a known probability density function ρ(x) is sampled by transforming a uniform variate by a transformation equation. The transformation equation is obtained by integrating ρ(x) and inverting the integral. If instead the transformation equation is given, one can reverse these steps to determine the probability density of a transformed uniform variate.

Generate a uniform variate x on the interval [0,1] and transform it according to either of the transformation equations below. In each case, plot a histogram of the resulting variate; determine an analytic form for the distribution, and plot the analytic function on top of the data to check your answer:

$$f(x) = x_{min}(x_{max}/x_{min})^x$$

for xmax > xmin > 0 (you can take xmin = 1 and xmax = 10);

$$f(x) = \sqrt{M^2 + M\Gamma \tan(M\Gamma x)}$$

for M > 0 and Γ > 0 (you can take M = 100 and Γ = 10).

Method

If x is uniform on [0, 1] and y = f(x) is increasing, ρ(y) dy = dx, so the density of y is ρ(y) = dx/dy. First transformation. Taking the logarithm and solving for x,

$$y = x_{min}\left(\frac{x_{max}}{x_{min}}\right)^x \implies x = \frac{\ln\frac{y}{x_{min}}}{\ln\frac{x_{max}}{x_{min}}}$$

so

$$\rho(y) = \frac{dx}{dy} = \frac{1}{y \ln x_{max}/x_{min}}, \quad x_{min} \le y \le x_{max}$$

With xmin = 1 and xmax = 10 this is

$$\rho(y) = \frac{1}{y\ \ln 10}, \quad 1 \le y \le 10$$

Second transformation. The argument MΓx = 1000x covers about 318 periods of the tangent, so the phase is effectively uniform and t = tan(MΓx) follows the Cauchy distribution:

$$t = \tan(M\Gamma x), \quad \rho(t) = \frac{1}{\pi(1 + t^2)}$$

Working with s = y^2,

$$s = y^2 = M^2 + M\Gamma t \implies \rho(s) = \frac{1}{\pi}\frac{M\Gamma}{(s - M^2)^2 + M^2\Gamma^2}$$

Changing variables to $y = \sqrt{s}$, with $ds/dy = 2y$, gives the following distribution, which peaks at $y \approx M$ with a full width of about Γ:

$$\rho(y) = 2y\,\rho(y^2) = \frac{2y}{\pi}\frac{M\Gamma}{(y^2 - M^2)^2 + M^2\Gamma^2}$$

The transformation is only real when $s \geq 0$. The fraction of uniform variates for which $M^2 + M\Gamma \tan(M\Gamma x) < 0$ is

$$P(s < 0) = P\left(t < -\frac{M}{\Gamma}\right) = \frac{\frac{\pi}{2} - \arctan(M/\Gamma)}{\pi} \approx 0.0317$$

$\rho(y)$ above integrates to $1 - 0.0317$ over $y > 0$.

Solution

I wrote the two analytic densities as functions, promoted them to F1D objects, and plotted each one in its own PlotView, placed side by side in a QSplitter. A helper function sets the title and axis labels of each view. One std::uniform_real_distribution on [0, 1] produces x, and every x is pushed through both transformations, with N = 10000000.

Both histograms have 100 bins (on [1, 10] and [50, 150]) and each entry has weight

$$w = \frac{1}{N\,\Delta y}$$

so, the histogram height is directly comparable with $\rho(y)$.

Result

